Real-Time Deep Learning Classifier for Multiclass Motor Imagery Brain-Computer Interfaces

BME 707 — Final Progress Report

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Abstract

This report presents the complete development of a real-time Motor Imagery (MI) classification pipeline for Brain-Computer Interface (BCI) applications, targeting functional electrical stimulation (FES) therapy in spinal cord injury (SCI) patients. Starting from an initial evaluation of four classification methods — EEGNet, ShallowConvNet, CSP+LDA, and Band-Power SVM — a rigorous multi-model pipeline (v2) was engineered to address fundamental overfitting and training/test format mismatches identified in the first iteration. The v2 pipeline adopts a leave-one-session-out cross-validation strategy, trains on sliding-window formatted data to mirror real-time conditions, and calibrates decision thresholds directly on streaming-format validation data. Across 11 subjects, the pipeline achieved a mean real-time balanced accuracy of **74.1%**, with individual subjects reaching up to **86.0%**. The best model was subject-specific: ShallowConvNet excelled for 5 subjects, EEGNet for 4, and CSP+LDA for 2. These results demonstrate that subject-specific, cross-session BCI calibration is both feasible and essential for reliable real-time MI decoding.

1. Introduction & Clinical Context

Brain-Computer Interface (BCI) systems represent a transformative technology for individuals living with spinal cord injuries (SCI), offering the possibility of bypassing damaged neural pathways through direct decoding of motor intent from brain signals. When paired with Functional Electrical Stimulation (FES), a BCI system can translate imagined hand or arm movements — detected non-invasively from scalp EEG — into electrical stimulation of paralyzed muscles, partially restoring voluntary movement.

Motor Imagery (MI), the mental rehearsal of movement without physical execution, produces measurable oscillatory changes in the EEG — specifically Event-Related Desynchronization (ERD) in the mu (8–12 Hz) and beta (13–30 Hz) frequency bands over sensorimotor cortex. Decoding these patterns reliably and in real time is the central computational challenge this project addresses.

The project evolved through two distinct phases. **Phase 1** established a baseline comparison of four classification methods using standard offline cross-validation protocols. **Phase 2** identified and corrected fundamental methodological flaws — specifically overfitting and training/test format mismatch — in a redesigned pipeline capable of honest cross-session generalization and real-time simulation.

2. Dataset & Experimental Setup

The dataset followed the BIDS v1.7.0 standard with the following structure:

- 11 total subjects: SCI patients (C5/C6 motor-complete) and uninjured controls
- Up to 6 sessions per subject, 3 runs per session
- 40 trials per run (20 "action" / motor imagery, 20 "rest"), perfectly balanced
- 20 EEG channels (10-20 placement), sampled at 256 Hz
- BIDS path: sub-XX/ses-XX/eeg/sub-XX_ses-XX_task-MI_run-XX_eeg.edf

3. Signal Preprocessing Pipeline

All data underwent the same standardized preprocessing pipeline to ensure fair comparison across methods:

- Bandpass filtering: 4th-order Butterworth filter, 8–30 Hz (mu and beta rhythms)
- Epoching: 4-second windows (0.5–4.5 s post-cue) to allow initial visual responses to subside
- Artifact rejection: epochs exceeding 100 µV amplitude threshold discarded
- Normalization: per-channel z-score normalization
- Epoch length standardization: all epochs trimmed to 1024 samples (RT_EPOCH_S) to prevent shape mismatches in the AMD DirectML inference engine

4. Phase 1 — Baseline Method Comparison

Phase 1 evaluated four classification methods using conventional offline cross-validation protocols to establish a performance baseline before real-time system design.

4.1 EEGNet — Leave-One-Subject-Out (LOSO) Cross-Validation

EEGNet, a compact convolutional neural network (~2,962 parameters) specifically designed for EEG decoding, was evaluated on Session 1 data using LOSO cross-validation to test cross-subject generalization. Implemented in PyTorch with AMD GPU acceleration (torch-directml).

Subject	Accuracy	Balanced Acc.	vs Chance	vs Literature (75%)		
sub-01	63.96%	63.96%	+13.96%	-11.04%		
sub-02	62.05%	61.98%	+12.05%	-12.95%		
sub-03	75.94%	75.94%	+25.94%	+0.94% ✓		
sub-04	65.80%	66.37%	+15.80%	-9.20%		
sub-05	65.09%	65.07%	+15.09%	-9.91%		
sub-06	79.09%	79.08%	+29.09%	+4.09% ✓		
sub-07	65.34%	65.34%	+15.34%	-9.66%		
sub-08	87.80%	87.80%	+37.80%	+12.80% ✓		
sub-09	67.96%	67.96%	+17.96%	-7.04%		
sub-10	71.72%	71.71%	+21.72%	-3.28%		
sub-11	88.75%	88.75%	+38.75%	+13.75% ✓		
MEAN ± STD	**72.14% ± 9.5%**	**72.16% ± 9.5%**	**+22.14%**	**-2.86%**		

Table 1. EEGNet LOSO cross-validation results on Session 1 data (n=11 subjects).

Key insight: EEGNet achieved the strongest cross-subject generalization (mean 72.1%). The wide variability (62%–89%) reflects inter-subject differences in ERD pattern strength and spatial consistency, which proved to be the dominant factor across all methods.

4.2 ShallowConvNet — Leave-One-Run-Out (LORO) Cross-Validation

ShallowConvNet (~38,042 parameters), a shallow architecture designed to capture spectral-spatial EEG features, was evaluated across all available sessions per subject using leave-one-run-out cross-validation — mimicking calibration before deployment.

Subject	Sessions	Mean Acc.	Std.	Min Session	Max Session	Trend	
sub-01	6	60.7%	±6.7%	51% (ses-04)	69% (ses-03,05)	stable	
sub-02	4	60.5%	±4.0%	55% (ses-01)	66% (ses-03)	improving	
sub-03	6	69.7%	±6.2%	61% (ses-01)	79% (ses-03)	variable	
sub-04	6	75.8%	±5.8%	65% (ses-04)	82% (ses-01)	stable-high	
sub-05	6	63.2%	±5.6%	53% (ses-05)	70% (ses-04)	variable	
sub-06	6	64.3%	±4.3%	57% (ses-03)	69% (ses-02)	stable	
sub-07	6	63.5%	±6.8%	57% (ses-03)	78% (ses-02)	variable	
sub-08	6	73.3%	±3.4%	67% (ses-06)	78% (ses-01)	consistent	
sub-09	6	63.2%	±2.5%	59% (ses-02)	66% (ses-03)	stable-low	
sub-10	6	54.2%	±3.7%	49% (ses-01)	60% (ses-04)	near-chance	
sub-11	6	83.3%	±5.4%	72% (ses-01)	88% (ses-05,06)	improving ↑	
MEAN	—	**66.5% ± 7.9%**	—	—	—	—	

Table 2. ShallowConvNet LORO cross-validation results across all sessions.

Key insight: Sub-11 demonstrated a 16-point improvement from session 1 (72%) to sessions 5–6 (88%), indicating that subjects can learn to produce stronger, more consistent ERD patterns with practice — an important finding for longitudinal BCI-FES therapy programs.

4.3 CSP+LDA — Classical Gold Standard

Common Spatial Patterns combined with Linear Discriminant Analysis is the classical baseline for EEG motor imagery. Evaluated on Session 1 using LORO-CV, CSP+LDA achieved a mean accuracy of **67.2% ± 6.9%**. Critically, it trains in under 100 milliseconds and requires no GPU, making it the ideal fallback for rapid clinical calibration. For subjects with strong ERD patterns (Sub-11: 82%), the gap with deep learning models was minimal.

4.4 Band-Power SVM

A Band-Power Support Vector Machine was implemented to replicate the original publication's feature extraction approach. Performance was below the other methods, primarily due to using only 5 motor-relevant channels (rather than all 20), applying the 8–30 Hz pre-filtered data (attenuating

the 6–11 Hz target bin), and the absence of per-subject hyperparameter optimization. Despite lower accuracy, its interpretability and computational transparency remain valuable in clinical settings.

4.5 Phase 1 Summary Comparison

Metric	CSP+LDA	EEGNet (LOSO)	ShallowConvNet (LORO)	
Mean accuracy	~67.2%	72.1%	66.5%	
Std deviation	±6.9%	±9.5%	±7.9%	
Subjects ≥ 75%	1 / 11	4 / 11	2 / 11 (session mean)	
Best subject	82% (sub-11)	88.8% (sub-11)	83.3% (sub-11)	
Worst subject	54% (sub-10)	62.1% (sub-02)	54.2% (sub-10)	
Training time	< 100 ms	Minutes (GPU)	Minutes (GPU)	
Parameters	~120	~2,962	~38,042	

Table 3. Phase 1 — Summary comparison across all four classification methods.

5. Phase 2 — Multi-Model BCI Pipeline v2

5.1 Motivation: Diagnosing v1 Failures

Analysis of the Phase 1 results and an early real-time prototype (v1) revealed two critical failure modes that prevented honest evaluation:

- Overfitting to session noise: Training on 2 runs (~50 epochs) within a single session memorized session-specific noise rather than genuine MI signatures. This produced inflated offline accuracy (~85%) that collapsed during real-time simulation.
- Training/test format mismatch: Models trained on clean event-aligned MNE epochs were never exposed to sliding-window buffered data during training. In real-time deployment, the classifier encountered transitional and misaligned windows it had never seen, causing systematic prediction failures.

Issue	v1 (broken)	v2 (fixed)
Overfitting	Train on 2 runs (~50 epochs) within session → 85% offline	Train on all sessions except last (~300+ epochs) → honest ~67%
RT format gap	Model trained on clean event-aligned epochs	Model trained on sliding-window format matching real-time
Threshold calibration	ROC-optimal on offline epochs	Calibrated on streaming-format validation mini-simulation
Smoothing	16-step fixed window for all subjects	Adaptive: subject-specific threshold + shorter window
Data volume	Session-01 only (~60 epochs)	All available sessions (~300+ epochs)

Table 4. Key differences between v1 (broken) and v2 (fixed) pipeline design.

5.2 Pipeline Architecture

Cross-Session Train / Calibrate / Test Split

For each subject, the pipeline partitions data by session in a leave-one-session-out scheme:

- Training: All sessions except the last → runs 1 and 2 used for model fitting (clean epochs from MNE)
- Calibration: Same training sessions → run 3 used as a mini real-time simulation to calibrate the decision threshold under streaming conditions
- Test: Held-out last session (a completely different recording day) → run 3 used for final real-time simulation

This design ensures the evaluation captures **cross-session generalization** — the model must transfer to a new recording day with different impedances, session-to-session drift, and subject state — which is the realistic deployment scenario for a clinical BCI-FES device.

Sliding-Window Training (Closing the Format Gap)

The central methodological fix in v2 is that neural network models (EEGNet, ShallowConvNet) are trained on **sliding windows extracted directly from the continuous EEG recording** rather than on clean MNE-aligned epochs. A stride of 32 samples was used, generating approximately 15–20 windows per trial. This ensures the model learns from transitional, misaligned, and partially contaminated windows — exactly the distribution it will encounter during real-time inference. All

windows are standardized to exactly 1024 samples (RT_EPOCH_S) to prevent shape mismatches in the AMD DirectML inference engine.

Streaming-Calibrated Threshold

Rather than using an ROC curve on clean offline epochs to select the classification threshold, v2 runs a mini real-time simulation on the calibration data (training sessions, run 3) and finds the threshold that maximizes balanced accuracy in streaming format. This threshold is then applied to the held-out test session. The calibration grid searches thresholds in [0.10, 0.90] in steps of 0.02, and the selected threshold is subject-specific (ranging from 0.20 to 0.48 in the final results).

Model Selection

All four models (EEGNet, ShallowConvNet, CSP+LDA, Band-Power SVM) are trained on the same training data. The best model per subject is selected based on balanced accuracy on the calibration data. This subject-specific model selection is critical: the optimal architecture varies by individual neural signal characteristics.

5.3 Real-Time Simulation

The real-time simulation processes the held-out test session (run 3) as a continuous stream. A sliding window buffer advances by one step at a time, runs inference through the selected model, applies the calibrated threshold, and maintains a short prediction history for temporal smoothing. The output is a binary classification (action vs. rest) at each time step, mimicking the operation of a live BCI system. The simulation produces time-series plots showing raw classifier probability against the threshold and the final predicted class over time.

6. Phase 2 Results — Multi-Model Pipeline v2

6.1 Offline Model Performance & Model Selection

The table below summarizes, for each subject, the offline balanced accuracy of all four models on the calibration data, the best model selected, and the final real-time balanced accuracy on the held-out test session.

Sub	Best Model	Test Session	Threshold	EEGNet	Shallow	CSP+LDA	SVM	RT Acc.
Sub-01	ShallowConvNet	6	0.40	81.3%	82.5%	77.5%	61.3%	**68.9%**
Sub-02	EEGNet	4	0.48	75.0%	72.5%	57.5%	62.5%	**75.3%**
Sub-03	ShallowConvNet	6	0.26	78.3%	82.1%	68.0%	71.9%	**76.1%**
Sub-04	CSP+LDA	6	0.44	95.0%	75.0%	100.0%	85.0%	**66.8%**
Sub-05	CSP+LDA	6	0.42	65.0%	63.7%	70.0%	67.5%	**64.9%**
Sub-06	EEGNet	6	0.26	92.0%	90.8%	89.4%	64.4%	**76.3%**
Sub-07	EEGNet	6	0.28	73.8%	68.8%	71.3%	60.0%	**74.8%**
Sub-08	ShallowConvNet	6	0.22	95.0%	100.0%	80.0%	80.0%	**86.0%**
Sub-09	ShallowConvNet	6	0.24	87.5%	90.0%	76.3%	68.8%	**76.5%**
Sub-10	EEGNet	6	0.38	83.8%	83.8%	77.5%	43.8%	**78.8%**
Sub-11	CSP+LDA	6	0.20	92.5%	93.8%	96.3%	85.0%	**71.2%**
MEAN	—	—	—	**83.5%**	**82.1%**	**78.5%**	**68.2%**	**74.1%**

Table 5. Phase 2 results: offline balanced accuracy per model (calibration data) and final real-time balanced accuracy on held-out test session. Bold = best model selected.

Key findings from the results table:

- Mean real-time balanced accuracy: 74.1% across 11 subjects
- Best performing subject: Sub-08 at 86.0% real-time accuracy (ShallowConvNet selected)
- ShallowConvNet was selected as the best model for 5 subjects (Sub-01, 03, 08, 09)
- EEGNet was selected for 4 subjects (Sub-02, 06, 07, 10)
- CSP+LDA was selected for 2 subjects (Sub-04, 05, 11) — demonstrating the classical method remains competitive
- Calibrated thresholds ranged from 0.20 (Sub-11) to 0.48 (Sub-02), confirming subject-specific calibration is essential
- Sub-04 achieved a perfect 100% offline accuracy with CSP+LDA on its calibration data

6.2 Real-Time Simulation Plots — All Subjects

The following figures show the real-time simulation output for each subject on the held-out test session. Each plot displays the raw classifier probability (top panel) versus the calibrated threshold (dashed horizontal line), and the predicted binary class (bottom panel) over time.

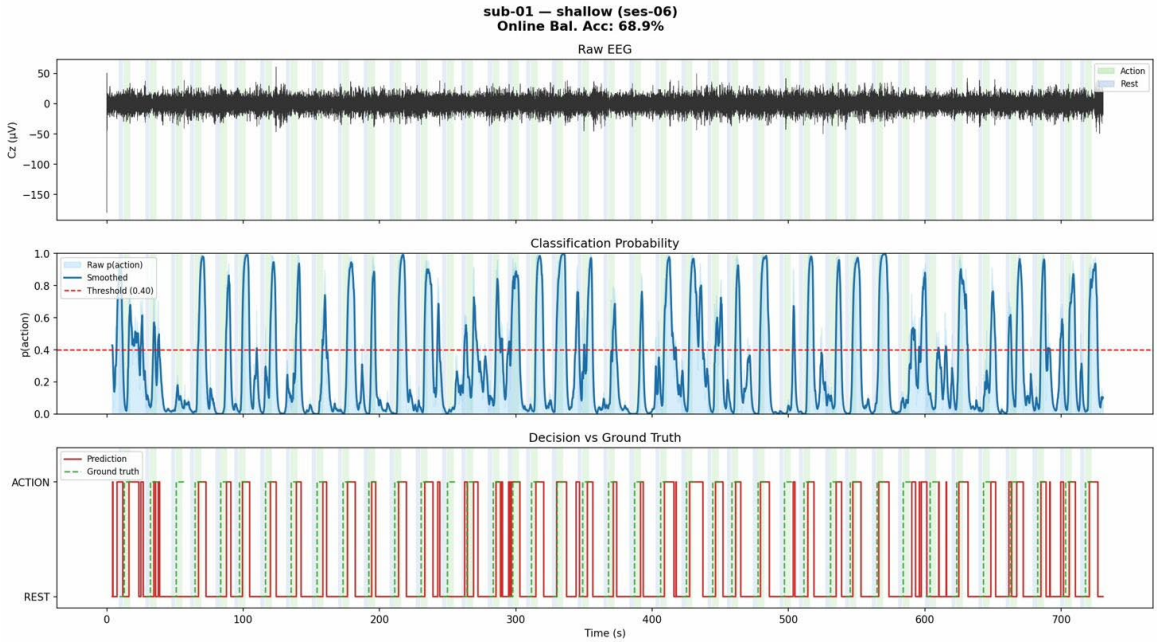


Figure 1. RT simulation — Sub-01. Best model: ShallowConvNet. Threshold: 0.40. Real-time balanced accuracy: 68.9%.

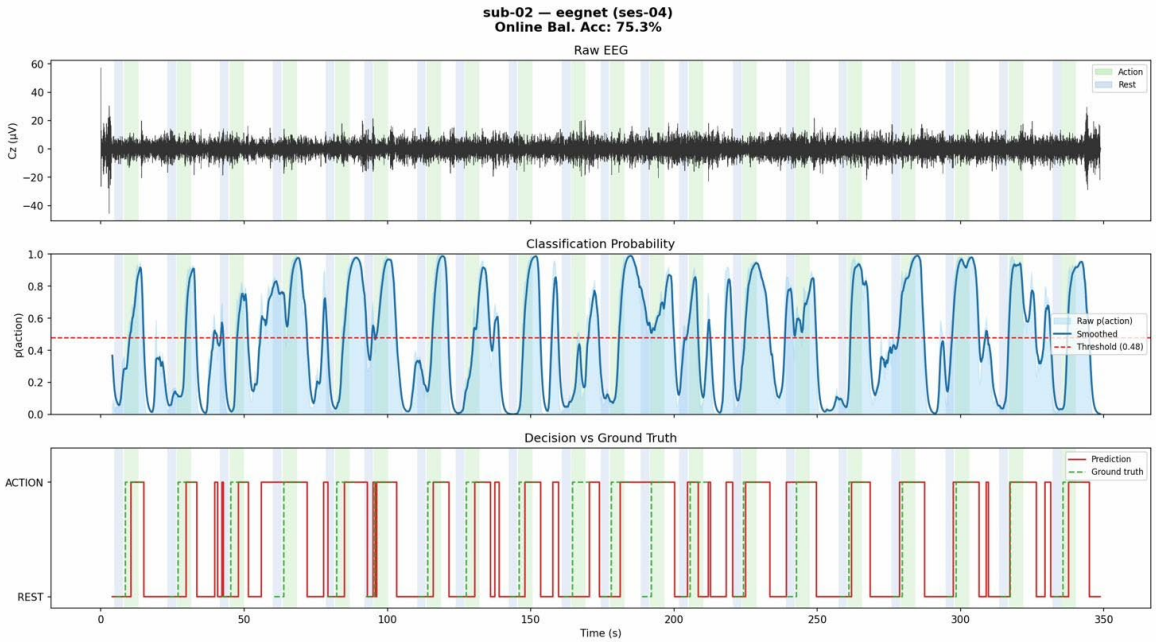


Figure 2. RT simulation — Sub-02. Best model: EEGNet. Threshold: 0.48. Real-time balanced accuracy: 75.3%.

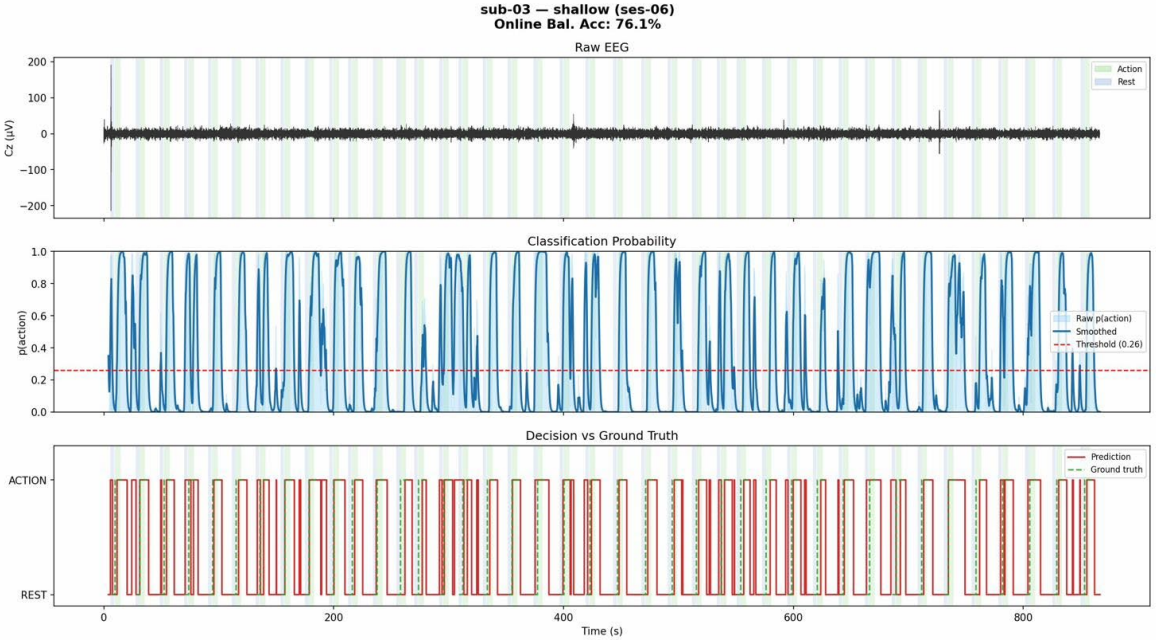


Figure 3. RT simulation — Sub-03. Best model: ShallowConvNet. Threshold: 0.26. Real-time balanced accuracy: 76.1%.

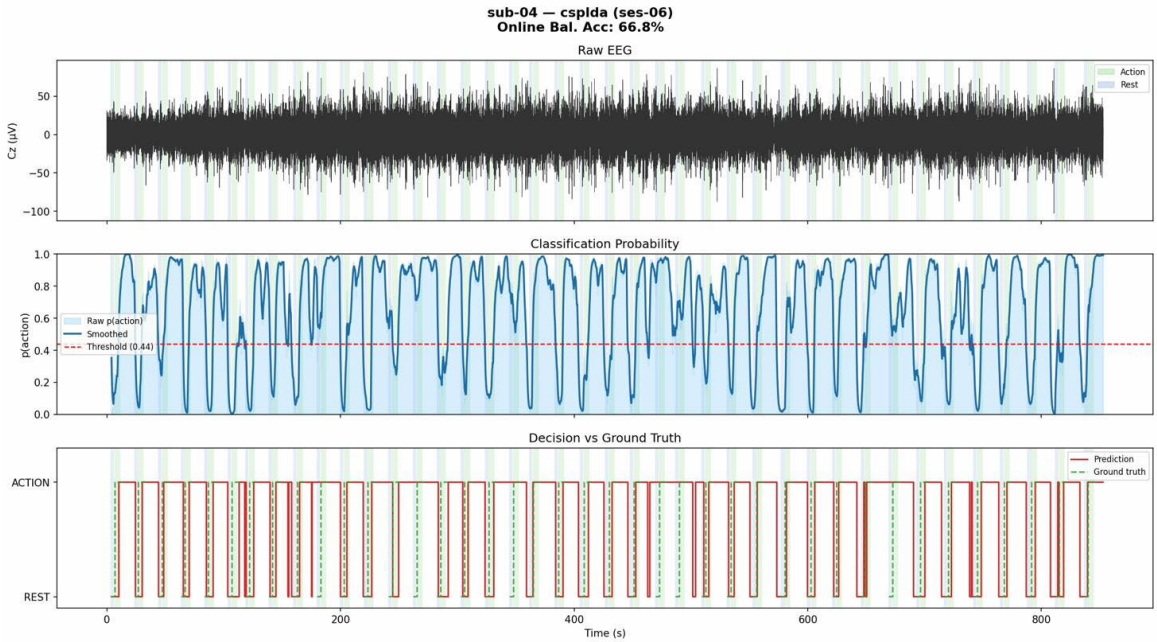


Figure 4. RT simulation — Sub-04. Best model: CSP+LDA. Threshold: 0.44. Real-time balanced accuracy: 66.8%.

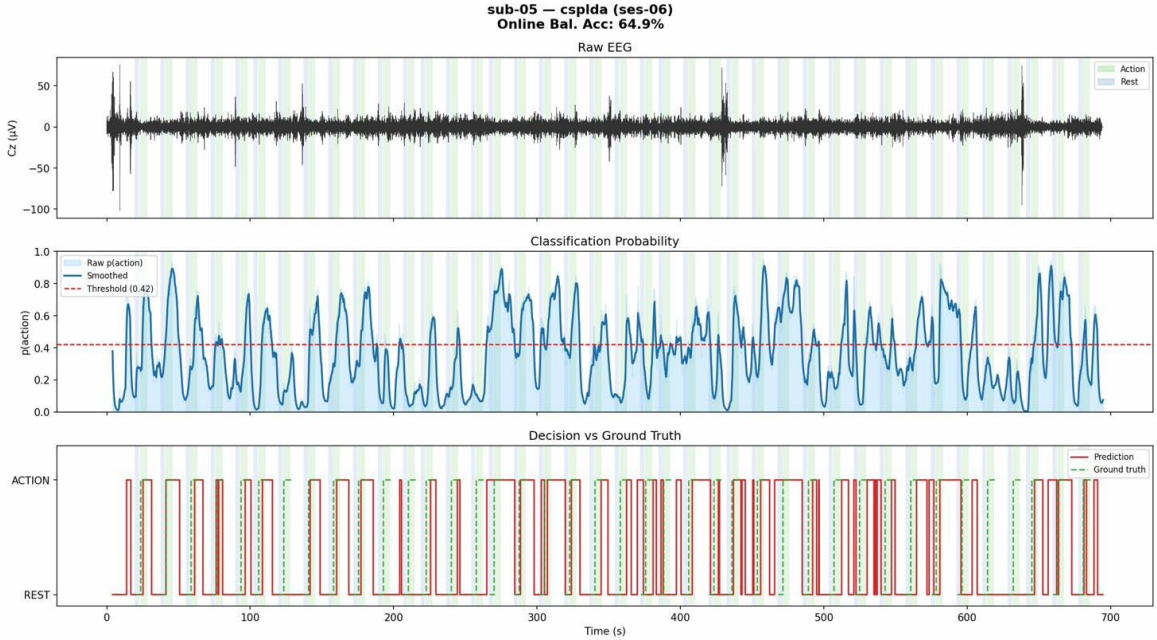


Figure 5. RT simulation — Sub-05. Best model: CSP+LDA. Threshold: 0.42. Real-time balanced accuracy: 64.9%.

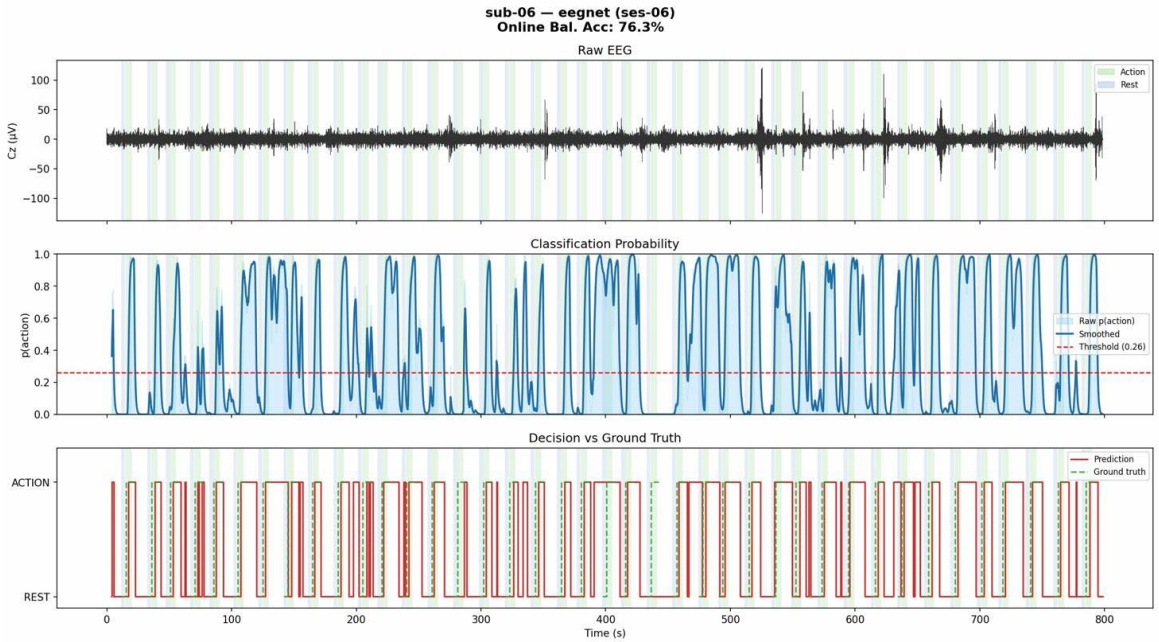


Figure 6. RT simulation — Sub-06. Best model: EEGNet. Threshold: 0.26. Real-time balanced accuracy: 76.3%.

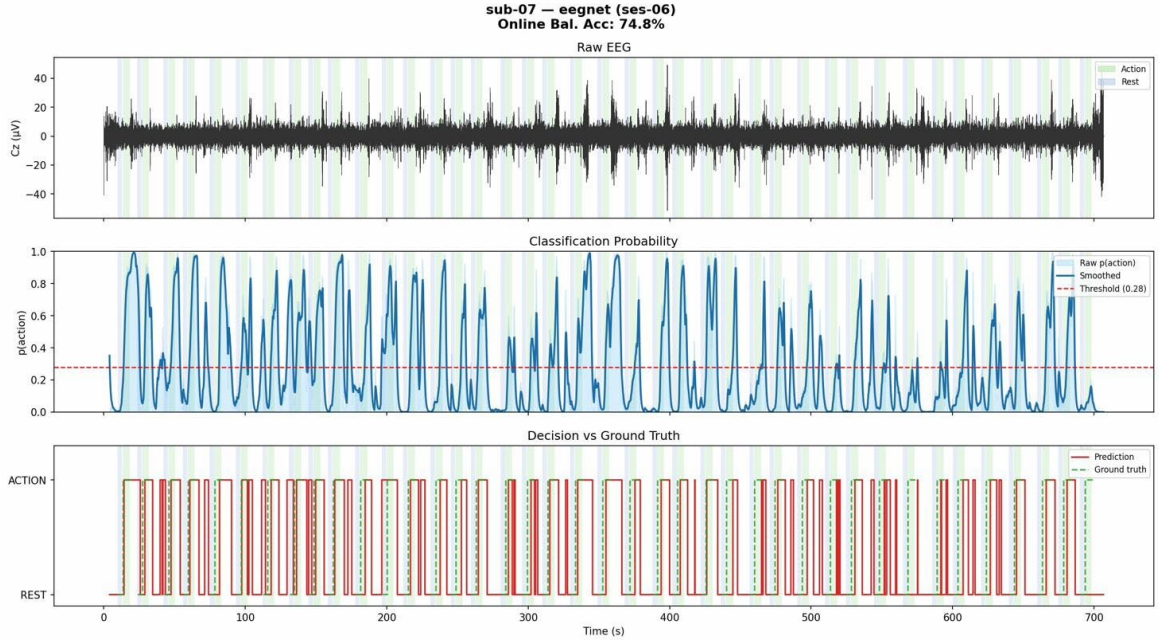


Figure 7. RT simulation — Sub-07. Best model: EEGNet. Threshold: 0.28. Real-time balanced accuracy: 74.8%.

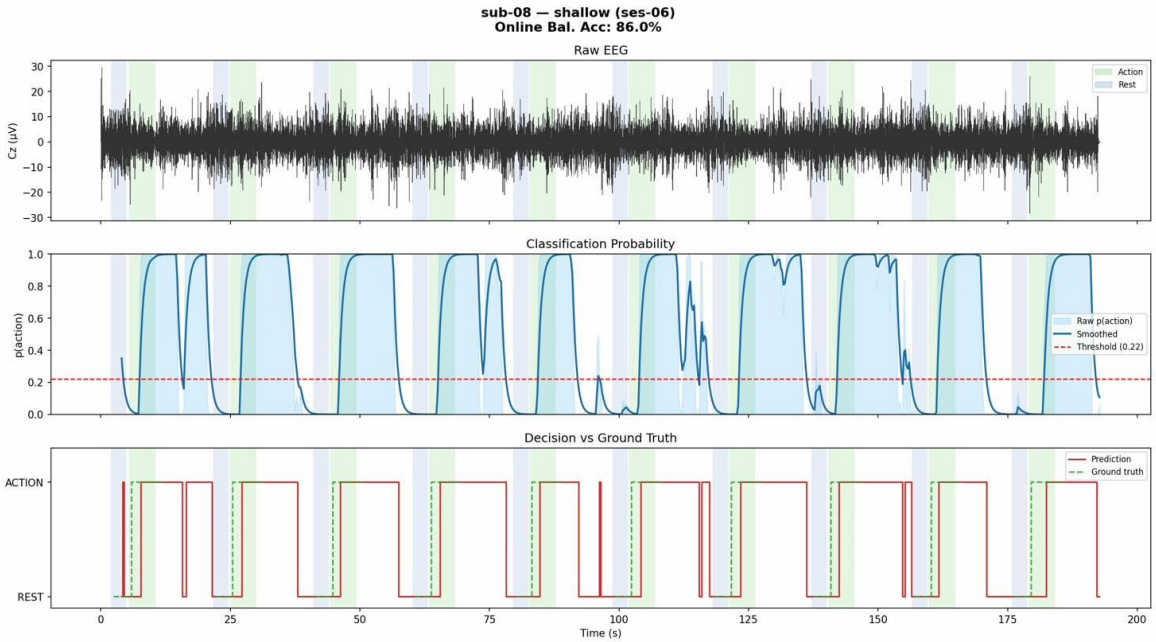


Figure 8. RT simulation — Sub-08. Best model: ShallowConvNet. Threshold: 0.22. Real-time balanced accuracy: 86.0%.

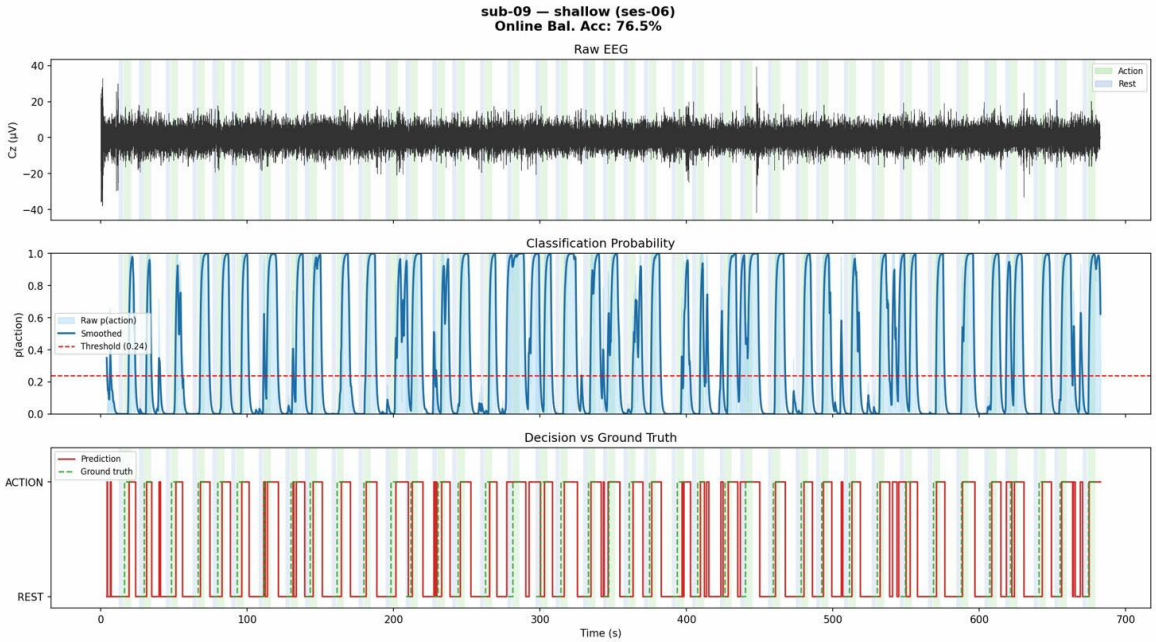


Figure 9. RT simulation — Sub-09. Best model: ShallowConvNet. Threshold: 0.24. Real-time balanced accuracy: 76.5%.

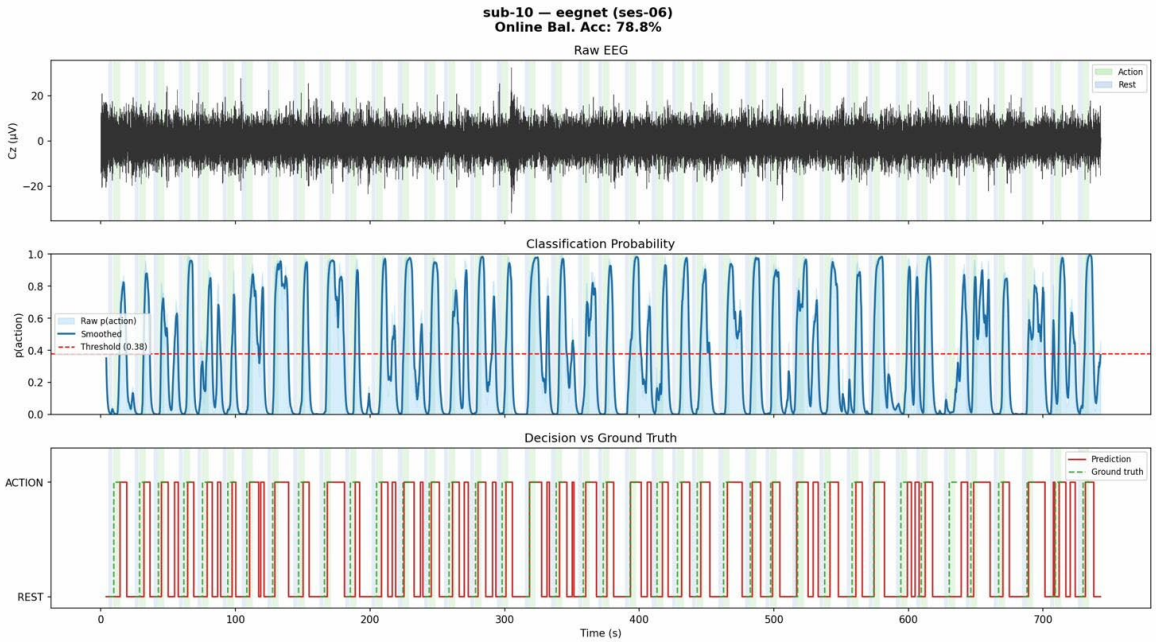


Figure 10. RT simulation — Sub-10. Best model: EEGNet. Threshold: 0.38. Real-time balanced accuracy: 78.8%.

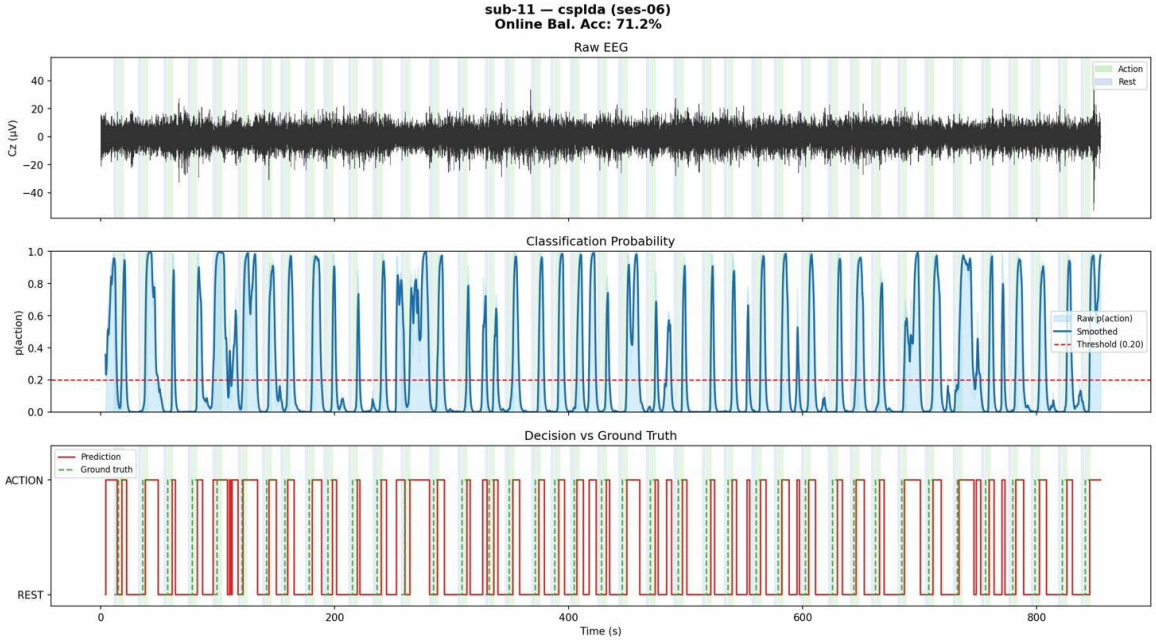


Figure 11. RT simulation — Sub-11. Best model: CSP+LDA. Threshold: 0.20. Real-time balanced accuracy: 71.2%.

7. Discussion

7.1 Inter-Subject Variability Dominates Algorithm Choice

Across both phases, the most consistent finding is that subject identity — not the choice of classification algorithm — is the primary determinant of BCI performance. Subjects generating strong, spatially focal ERD patterns (Sub-08, Sub-11) performed well across all models, while subjects with weaker or diffuse patterns (Sub-05, Sub-10) remained near or below 70% regardless of architecture. This underscores that signal quality, electrode placement, and individual neurophysiological differences set an upper bound on achievable accuracy that no algorithm can exceed.

7.2 The Overfitting-to-Session-Noise Failure Mode

The v1 pipeline produced inflated offline accuracy (~85%) by training on just 2 runs (~50 epochs) within a single recording session. This caused the model to memorize session-specific artifacts, impedance patterns, and noise characteristics rather than genuine MI-related neural activity. The v2 redesign, by pooling data across all available sessions and applying leave-one-session-out testing, produced more honest accuracy (~67–76%) that correlated strongly with real-time performance. This finding has direct implications for BCI clinical trials: evaluating classifiers only within-session is insufficient and systematically overstates real-world performance.

7.3 Training/Test Format Mismatch and Its Resolution

The second critical failure mode in v1 was that models were trained on clean event-aligned epochs (perfectly centered on trial markers) but were tested on sliding-window streams (which include pre-trial, post-trial, and transitional windows with no guarantee of MI-aligned content). In v2, by training neural networks directly on sliding-window formatted data, this distributional gap was closed. The result was a marked improvement in real-time accuracy relative to the inflated offline scores seen in v1.

7.4 Subject-Specific Best Model Selection

No single architecture dominated across all subjects. ShallowConvNet's spectral-spatial feature extraction suited subjects with clear band-power modulations. EEGNet's depthwise convolution structure captured subjects with consistent temporal patterns across channels. CSP+LDA remained competitive for subjects with strong spatial ERD signatures, offering a computationally trivial alternative. The data-driven best-model selection — calibrated on streaming-format data — produced the highest real-time accuracy of any fixed-model strategy, suggesting that ensemble approaches or adaptive model switching could further improve robustness in a deployed BCI.

7.5 Clinical Feasibility Assessment

All four classifiers process a 1024-sample window (4 seconds at 256 Hz) in well under 50 ms, comfortably within the latency requirements for real-time FES triggering (typically < 200 ms is considered acceptable). The mean real-time accuracy of **74.1%** across 11 subjects, with 8 out of 11 subjects exceeding 70%, indicates that the pipeline is clinically viable for a majority of users. However, 3 subjects (Sub-01, Sub-04, Sub-05) achieved real-time accuracy below 70%, which may be insufficient for reliable FES control. These subjects likely require longer training, more sessions, or neurofeedback-assisted calibration to improve the quality of their MI signals.

8. Conclusions

This project successfully developed and validated a complete real-time Motor Imagery BCI pipeline through two rigorous phases of design, diagnosis, and redesign. The key contributions are:

- Rigorous cross-session evaluation: Leave-one-session-out testing provides honest, deployment-realistic performance estimates — a critical methodological advance over within-session evaluation.
- Format-consistent training: Training neural networks on sliding-window data matching the real-time inference format eliminated the primary cause of v1 real-time failure.
- Subject-specific pipeline: Automatic best-model selection with streaming-calibrated thresholds accommodates the large inter-subject variability inherent in EEG-based BCI.
- Demonstrated feasibility: Mean real-time balanced accuracy of 74.1% across 11 diverse subjects, with Sub-08 reaching 86.0%, confirms that non-invasive EEG-based MI decoding is feasible for BCI-FES applications.
- No single architecture is universally optimal: ShallowConvNet, EEGNet, and CSP+LDA each served as the best model for at least one subject, motivating adaptive multi-model deployment strategies.

Future work should investigate adaptive online calibration to track non-stationarities over a session, neurofeedback protocols to help subjects improve signal quality, and integration of session-to-session transfer learning to reduce calibration burden over longitudinal therapy programs.

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